

Extracting the Rapidity Divergence from Leading-Order Trijet Production at Small x

Phase-space slicing of the $q\bar{q}g$ real cross-section, lifetime ordering,
and the matching onto B-JIMWLK / BK evolution

Working notes. Conventions and all quoted explicit results follow P. Caucal, F. Salazar, B. Schenke, R. Venugopalan, *Back-to-back inclusive dijets in DIS at small x : Sudakov suppression and gluon saturation at NLO*, arXiv:2208.13872v2 [CSSV]; the slicing machinery follows B. W. Harris and

J. F. Owens, *Two cutoff phase space slicing method*, Phys. Rev. D **65** (2002) 094032 [HO].

Equation numbers of the form (3.6), (B.7), (4.51) refer to CSSV; those of the form HO (2.16) refer to Harris–Owens.

0. The answer, in one page

You have the leading-order **trijet** (three-parton, $q\bar{q}g$) cross-section in the CGC. Integrated over the gluon phase space it is **not** finite: it carries one logarithmic **rapidity** (slow-gluon) divergence, one **soft** double logarithm, and one **final-state collinear** pole. The goal is to peel these off with a phase-space slice and be left with (i) a finite, fully differential, three-parton weight you can integrate by Monte Carlo, (ii) a finite two-parton (dijet) weight that you add to the virtual corrections, and (iii) one single-logarithmic piece — and *only* that piece — which is absorbed into B-JIMWLK (large N_c : BK) evolution of the target.

Master decomposition. With δ the soft slicing parameter, R the jet radius (collinear slicing parameter), $z_f = k_f^-/q^-$ the rapidity factorization scale, $Q_f^2 = 2k_f^+k_f^-$ its conjugate hard scale, and $z_0 = \Lambda^-/q^-$ the rapidity regulator:

$$\underbrace{\int_{\Phi_g} d\sigma_3^{\text{LO}} \mathcal{M}_2}_{\text{trijet, divergent}} + \underbrace{d\sigma_V}_{\text{dijet, divergent}} = \underbrace{\ln\left(\frac{z_f}{z_0}\right) \mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}}}_{\text{(I) B-JIMWLK / BK}} + \underbrace{\Delta_{\text{lt}}(Q_f)}_{\text{(II) lifetime add-back}} + \underbrace{d\sigma_{\text{fin}}^{(2)}(\delta, R, z_f)}_{\text{(III) finite dijet}} + \underbrace{\int_{\text{HC}} d\sigma_3^{\text{LO}} \mathcal{M}_3}_{\text{(IV) finite trijet}}$$

Every one of (II), (III), (IV) is finite in $d = 2$ transverse dimensions and free of $\ln(1/z_0)$. Piece (I) is the *entire* divergence and is the only thing that renormalizes the target. The sum of (III) + (IV) is independent of the arbitrary slicing parameter δ ; the sum of (I) + (II) + (III) + (IV) is independent of z_f and Q_f up to $\mathcal{O}(\alpha_s^2)$.

The one non-obvious ingredient is piece (II). If you define the "slow" region simply as $z_g < z_f$ — a straight cut in the gluon's minus-momentum fraction — you **over-subtract**. The naive slow limit is only legitimate where the emitted gluon's lifetime is shorter than the parent $q\bar{q}$ dipole's lifetime,

$$\frac{1}{k_g^+} \leq \frac{1}{k_f^+}, \quad k_g^+ = \frac{k_{g\perp}^2}{2z_g q^-},$$

and the region you wrongly removed must be given back. Adding it back produces exactly

$$\Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi} \ln^2(Q_f^2 r_{b\bar{b}'}^2) \times (\text{LO}),$$

a **negative** Sudakov double logarithm. Getting the sign of that double log right is the sharpest diagnostic that your divergence extraction is correct: a naive $z_g < z_f$ slice gives it with the wrong (positive) sign, which is unphysical. This is the content of §4.3 of CSSV and it is the reason the lifetime-ordering paper is relevant to your problem.

1. Setup, frame and notation

1.1 Kinematics

Work in the dipole (projectile) frame. The virtual photon carries a large minus light-cone momentum and no transverse momentum; the nucleus carries a large plus momentum:

$$q^\mu = \left(-\frac{Q^2}{2q^-}, q^-, \mathbf{0}_\perp\right), \quad P^\mu = \left(P^+, \frac{M^2}{2P^+}, \mathbf{0}_\perp\right), \quad a \cdot b = a^+ b^- + a^- b^+ - \mathbf{a}_\perp \cdot \mathbf{b}_\perp.$$

Photon polarization $\lambda = 0$ (L), ± 1 (T); $x_{Bj} = Q^2/(2q^-P^+)$. The quark, antiquark and gluon four-momenta are $k_1^\mu, k_2^\mu, k_g^\mu$ with longitudinal momentum fractions

$$z_1 = \frac{k_1^-}{q^-}, \quad z_2 = \frac{k_2^-}{q^-}, \quad z_g = \frac{k_g^-}{q^-}.$$

At LO (dijet) $\delta_z^{(2)} = \delta(1 - z_1 - z_2)$; for the trijet $\delta_z^{(3)} = \delta(1 - z_1 - z_2 - z_g)$. Transverse positions at which the partons cross the shockwave: $\mathbf{x}_\perp$ (quark), $\mathbf{y}_\perp$ (antiquark), $\mathbf{z}_\perp$ (gluon); primed coordinates in the conjugate amplitude. Differences are written $\mathbf{r}_{xy} \equiv \mathbf{x}_\perp - \mathbf{y}_\perp$, and

$$d^8 \mathbf{X}_\perp = d^2 \mathbf{x}_\perp d^2 \mathbf{x}'_\perp d^2 \mathbf{y}_\perp d^2 \mathbf{y}'_\perp, \quad \bar{Q}^2 = z_1 z_2 Q^2.$$

Back-to-back variables (needed only for the Sudakov discussion):

$$\mathbf{q}_\perp = \mathbf{k}_{1\perp} + \mathbf{k}_{2\perp}, \quad \mathbf{P}_\perp = z_2 \mathbf{k}_{1\perp} - z_1 \mathbf{k}_{2\perp}, \quad \mathbf{u}_\perp = \mathbf{x}_\perp - \mathbf{y}_\perp, \quad \mathbf{b}_\perp = z_1 \mathbf{x}_\perp + z_2 \mathbf{y}_\perp,$$

with $|\mathbf{u}_\perp| \sim 1/|\mathbf{P}_\perp|$ and $|\mathbf{b}_\perp| \sim 1/|\mathbf{q}_\perp|$ conjugate pairs. The back-to-back limit is $|\mathbf{q}_\perp| \ll |\mathbf{P}_\perp|$.

1.2 The three rapidity scales — keep them distinct

Symbol	Meaning	Status
$z_0 = \Lambda^-/q^-$	Lower cutoff on the gluon's minus momentum; the <i>regulator</i> of the rapidity divergence	Unphysical. Must disappear from the final answer, absorbed into the initial condition of the evolution at $Y_0 = \ln(1/z_0)$.
$z_f = k_f^-/q^-$	Rapidity <i>factorization scale</i> ; splits "slow" ($\rightarrow$ evolution) from "fast" ($\rightarrow$ impact factor)	Arbitrary but physical-scale-like, analogous to μ_f in collinear factorization. Typical choices $z_f = \min(z_1, z_2)$ or $z_1 z_2$. Residual dependence is $\mathcal{O}(\alpha_s^2)$.
$k_f^+, Q_f^2 = 2k_f^+ k_f^-$	The <i>conjugate</i> plus-momentum scale that enters through lifetime ordering	Parametrically $Q_f^2 \sim \mathbf{P}_\perp^2$. Fixes the scheme of the single Sudakov logarithm; the double log is independent of it.
δ	Soft phase-space <i>slicing</i> parameter on z_g	Pure bookkeeping. Cancels exactly between the analytic 2-body weight and the numerical 3-body weight. Analogue of HO's δ_s .
R	Jet radius = collinear slicing parameter	Physical (part of the observable). Analogue of HO's δ_c , but not sent to zero: $\ln R$ survives in the answer. $\mathcal{O}(R^2)$ is dropped.

The single most common error in this calculation is to conflate z_0 , z_f and δ . $\ln(1/z_0)$ is a divergence and must leave the impact factor. $\ln(1/z_f)$ is a finite logarithm that *stays* in the impact factor and is what makes the impact factor factorization-scheme dependent. $\ln(1/\delta)$ must cancel identically.

1.3 Wilson lines and the colour correlators

Eikonal propagation through the shockwave is a Wilson line in the fundamental (quark) or adjoint (gluon) representation,

$$V(\mathbf{z}_\perp) = \mathcal{P} \exp \left(ig \int_{-\infty}^{\infty} dz^- A_{cl}^{+,c}(z^-, \mathbf{z}_\perp) t^c \right), \quad U(\mathbf{z}_\perp) = \mathcal{P} \exp \left(ig \int_{-\infty}^{\infty} dz^- A_{cl}^{+,c}(z^-, \mathbf{z}_\perp) T^c \right),$$

[CSSV (A.5)–(A.6)], with dipole and quadrupole operators

$$D_{xy} = \frac{1}{N_c} \text{Tr} (V(\mathbf{x}_\perp) V^\dagger(\mathbf{y}_\perp)), \quad Q_{xy,y'x'} = \frac{1}{N_c} \text{Tr} (V(\mathbf{x}_\perp) V^\dagger(\mathbf{y}_\perp) V(\mathbf{y}'_\perp) V^\dagger(\mathbf{x}'_\perp)).$$

All observables are CGC averages $\langle \mathcal{O} \rangle_Y = \int \mathcal{D}[\rho_A] W_Y[\rho_A] \mathcal{O}[\rho_A]$ over the colour-source weight functional at projectile rapidity Y . The five correlators that appear at NLO are CSSV Table 1:

$$\begin{aligned} \Xi_{\text{LO}} &= \langle 1 - D_{xy} - D_{y'x'} + Q_{xy,y'x'} \rangle, & \Xi_{\text{NLO},3} &= \frac{N_c}{2} \langle 1 - D_{xy} - D_{y'x'} + D_{xy} D_{y'x'} \rangle - \frac{1}{2N_c} \Xi_{\text{LO}}, \\ \Xi_{\text{NLO},1} &= \frac{N_c}{2} \langle 1 - D_{y'x'} + Q_{zy,y'x'} D_{xz} - D_{xz} D_{zy} \rangle - \frac{1}{2N_c} \Xi_{\text{LO}}, \\ \Xi_{\text{NLO},2} &= \frac{N_c}{2} \langle 1 - D_{y'x'} + Q_{xz,y'x'} D_{zy} - D_{xz} D_{zy} \rangle - \frac{1}{2N_c} \Xi_{\text{LO}}, \\ \Xi_{\text{NLO},4} &= \frac{N_c}{2} \langle Q_{z'x',xz} Q_{zy,y'z'} - D_{xz} D_{zy} - D_{y'z'} D_{z'x'} + 1 \rangle - \frac{1}{2N_c} \Xi_{\text{LO}}. \end{aligned}$$

$\Xi_{\text{NLO},1,2,4}$ depend on the gluon transverse coordinate $\mathbf{z}_\perp$ (the gluon crossed the shockwave); Ξ_{LO} and $\Xi_{\text{NLO},3}$ do not. That distinction is what organizes the divergence structure, as we now show.

1.4 The LO dijet cross-section

$$\left. \frac{d\sigma^{\gamma_\lambda^* + A \rightarrow q\bar{q} + X}}{d^2\mathbf{k}_{1\perp} d^2\mathbf{k}_{2\perp} d\eta_1 d\eta_2} \right|_{\text{LO}} = \frac{\alpha_{\text{em}} e_f^2 N_c \delta_z^{(2)}}{(2\pi)^6} \int d^8\mathbf{X}_\perp e^{-i\mathbf{k}_{1\perp} \cdot \mathbf{r}_{xx'}} e^{-i\mathbf{k}_{2\perp} \cdot \mathbf{r}_{yy'}} \Xi_{\text{LO}} \mathcal{R}_{\text{LO}}^\lambda(\mathbf{r}_{xy}, \mathbf{r}_{x'y'})$$

[CSSV (2.6)], with the QED splitting factors

$$\mathcal{R}_{\text{LO}}^L = 8z_1^3 z_2^3 Q^2 K_0(\bar{Q}r_{xy}) K_0(\bar{Q}r_{x'y'}), \quad \mathcal{R}_{\text{LO}}^T = 2z_1 z_2 [z_1^2 + z_2^2] \frac{\mathbf{r}_{xy} \cdot \mathbf{r}_{x'y'}}{r_{xy} r_{x'y'}} \bar{Q}^2 K_1(\bar{Q}r_{xy}) K_1(\bar{Q}r_{x'y'}).$$

This is completely finite. Every divergence discussed below is a property of the *trijet* (and of the one-loop dijet), not of this object.

1.5 The LO trijet: structure that matters for the divergences

For $\gamma_\lambda^* + A \rightarrow q(k_1) \bar{q}(k_2) g(k_g) + X$ the amplitude splits, in the shockwave approximation, into two classes according to **whether the gluon is emitted before or after the shockwave**:

$$\mathcal{M} = \mathcal{M}_{\text{bef}} + \mathcal{M}_{\text{aft}}.$$

- $\mathcal{M}_{\text{bef}}$ (**gluon crosses the shockwave**). Three Wilson lines, $V(\mathbf{x}_\perp) \otimes V^\dagger(\mathbf{y}_\perp) \otimes U(\mathbf{z}_\perp)$; these are the contributions that generate $\Xi_{\text{NLO},1}, \Xi_{\text{NLO},2}, \Xi_{\text{NLO},4}$ and the $\mathbf{z}_\perp$ integration. The light-cone wave function involves $K_0(QX_R)$ with the **effective $q\bar{q}g$ dipole size**

$$X_R^2 = z_1 z_2 \mathbf{r}_{xy}^2 + z_1 z_g \mathbf{r}_{zx}^2 + z_2 z_g \mathbf{r}_{zy}^2 \quad [\text{CSSV (3.25)}].$$

- $\mathcal{M}_{\text{aft}}$ (**gluon emitted after the shockwave; diagrams R_2, R'_2 of CSSV Fig. 1**). Only the $q\bar{q}$ dipole $V(\mathbf{x}_\perp) V^\dagger(\mathbf{y}_\perp)$ appears, so these terms carry Ξ_{LO} or $\Xi_{\text{NLO},3}$, i.e. *no $\mathbf{z}_\perp$ dependence*. The wave function is the LO one at shifted virtuality, $K_0(\bar{Q}_{R_2} r_{xy})$ with $\bar{Q}_{R_2}^2 = z_2(1 - z_2)Q^2$, multiplied by the **eikonal emission factor**

$$\frac{e^{-i\mathbf{k}_{g\perp} \cdot \mathbf{r}_{xx'}}}{(\mathbf{k}_{g\perp} - \frac{z_g}{z_1} \mathbf{k}_{1\perp})^2} \quad [\text{CSSV (B.1)}].$$

Why this split is the right organizing principle. Only $\mathcal{M}_{\text{aft}}$ has an on-shell internal quark propagator as $k_g^\mu \rightarrow 0$; hence *only* $R_2 \times R_2, R'_2 \times R'_2$ and $R_2 \times R'_2$ carry the genuine **soft** ($\ln^2$) divergence and the final-state **collinear** pole. The *rapidity* ($\ln$) divergence, on the other hand, is carried by **all** diagrams, because $\int dz_g/z_g$ comes from the phase-space measure. Keeping these two facts apart is the whole game.

The explicit trijet expressions are long; the pieces we actually need are CSSV (B.1) [$R_2 \times R_2$], (B.11) [$R_2 \times R'_2$], and (3.23) [gluon crossing the shockwave]. They are reproduced in §7.

2. Complete divergence inventory of the LO trijet

The gluon phase space measure carries an explicit $1/z_g$:

$$\int d\Phi_g = \int \frac{d\mathbf{k}_g^- d^2\mathbf{k}_{g\perp}}{(2\pi)^3 2k_g^-} = \frac{1}{2(2\pi)^3} \int_{z_0}^{1-z_1} \frac{dz_g}{z_g} \int d^2\mathbf{k}_{g\perp}.$$

Divergences arise wherever the squared amplitude fails to vanish fast enough at the boundaries of this integration. There are exactly three such places.

2.1 (D1) Rapidity ("slow-gluon") divergence — the one that goes into evolution

Limit: $z_g \rightarrow 0$ at fixed $\mathbf{k}_{g\perp}$ (equivalently, fixed $\mathbf{z}_\perp$). **Regulator:** $z_g > z_0 = \Lambda^-/q^-$. **Strength:** single logarithm $\ln(1/z_0)$.

In this limit the $q\bar{q}g$ light-cone wave functions collapse onto the LO one,

$$X_R^2 \xrightarrow{z_g \rightarrow 0} z_1 z_2 \mathbf{r}_{xy}^2, \quad X_V^2 \xrightarrow{z_g \rightarrow 0} z_1 z_2 \mathbf{r}_{xy}^2, \quad K_0(QX_{R/V}) \rightarrow K_0(\bar{Q}r_{xy}),$$

and what is left multiplying dz_g/z_g is a purely transverse emission kernel of dipole type. Using CSSV's identity (3.27),

$$\frac{1}{\pi} \int d^2 \mathbf{z}_\perp \left[\frac{\mathbf{r}_{zx} \cdot \mathbf{r}_{zx'}}{\mathbf{r}_{zx}^2 \mathbf{r}_{zx'}^2} - \frac{\mathbf{r}_{zx} \cdot \mathbf{r}_{zy}}{\mathbf{r}_{zx}^2 \mathbf{r}_{zy}^2} \right] = \ln \left(\frac{\mathbf{r}_{xy}^2}{\mathbf{r}_{xx'}^2} \right),$$

the slow limit of the LO colour structure organizes into [CSSV (3.28)]

$$d\sigma_{\text{LO,slow}} = \frac{\alpha_{\text{em}} e_f^2 \delta_z^{(2)}}{(2\pi)^6} \int d^8 \mathbf{X}_\perp \mathcal{R}_{\text{LO}}^\lambda e^{-i\mathbf{k}_{1\perp} \cdot \mathbf{r}_{xx'} - i\mathbf{k}_{2\perp} \cdot \mathbf{r}_{yy'}} \frac{\alpha_s C_F}{\pi^2} \ln \left(\frac{k_f^-}{\Lambda^-} \right) \int d^2 \mathbf{z}_\perp \mathcal{K}_{\text{dip}} \Xi_{\text{LO}},$$

$$\mathcal{K}_{\text{dip}} = \frac{\mathbf{r}_{zx} \cdot \mathbf{r}_{zx'}}{\mathbf{r}_{zx}^2 \mathbf{r}_{zx'}^2} - \frac{\mathbf{r}_{zx} \cdot \mathbf{r}_{zy}}{\mathbf{r}_{zx}^2 \mathbf{r}_{zy}^2} + \frac{\mathbf{r}_{zy} \cdot \mathbf{r}_{zy'}}{\mathbf{r}_{zy}^2 \mathbf{r}_{zy'}^2} - \frac{\mathbf{r}_{zx'} \cdot \mathbf{r}_{zy'}}{\mathbf{r}_{zx'}^2 \mathbf{r}_{zy'}^2}.$$

Collecting *all* colour channels (there are six independent transverse structures once the quadrupoles are included) gives the complete rapidity divergence, CSSV (3.29), which is exactly the leading-log JIMWLK Hamiltonian acting on the LO cross-section:

$$d\sigma_{\text{slow}} = \ln \left(\frac{k_f^-}{\Lambda^-} \right) \mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}}, \quad [\text{CSSV (3.32)}]$$

$$\mathcal{H}_{\text{LL}} = \frac{1}{2} \int d^2 \mathbf{u}_\perp d^2 \mathbf{v}_\perp \frac{\delta}{\delta A_{\text{cl}}^{+,a}(\mathbf{u}_\perp)} \eta^{ab}(\mathbf{u}_\perp, \mathbf{v}_\perp) \frac{\delta}{\delta A_{\text{cl}}^{+,b}(\mathbf{v}_\perp)},$$

$$\eta^{ab}(\mathbf{u}_\perp, \mathbf{v}_\perp) = \frac{1}{\pi} \int \frac{d^2 \mathbf{z}_\perp}{(2\pi)^2} \frac{(\mathbf{u}_\perp - \mathbf{z}_\perp) \cdot (\mathbf{v}_\perp - \mathbf{z}_\perp)}{(\mathbf{u}_\perp - \mathbf{z}_\perp)^2 (\mathbf{v}_\perp - \mathbf{z}_\perp)^2} [1 + U^\dagger(\mathbf{u}_\perp) U(\mathbf{v}_\perp) - U^\dagger(\mathbf{u}_\perp) U(\mathbf{z}_\perp) - U^\dagger(\mathbf{z}_\perp) U(\mathbf{v}_\perp)]^{ab}.$$

Crucial bookkeeping point. The four terms in the bracket of η^{ab} are *not* all supplied by the real trijet. The "1" and the $U^\dagger U$ terms come from the **virtual** $q\bar{q}$ diagrams in their own $z_g \rightarrow 0$ limit; the $U^\dagger(\mathbf{u})U(\mathbf{z})$ and $U^\dagger(\mathbf{z})U(\mathbf{v})$ terms come from the **real** trijet. You cannot build B-JIMWLK (nor BK) from the trijet alone: without the virtual slow pieces the kernel is not real-virtual complete, has no probabilistic interpretation, and the $\mathbf{z}_\perp$ integral does not converge. So "take the divergence out of the trijet" in practice means "take the divergence out of the trijet *and* out of the one-loop dijet, and recognize the sum as $\mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}}$ ".

2.2 (D2) Soft divergence — a double log, and it must cancel, not evolve

Limit: $k_g^\mu \rightarrow 0$, i.e. $z_g \rightarrow 0$ and $\mathbf{k}_{g\perp} \rightarrow 0$ together. **Origin:** only $\mathcal{M}_{\text{aft}}(R_2, R'_2)$, where the internal quark propagator goes on shell. **Strength:** $\ln^2(1/z_0)$.

Concretely, isolate from $R_2 \times R_2$ the strict eikonal piece [CSSV (B.4)]

$$d\sigma_{R_2 \times R_2, \text{soft-div}} \propto \Xi_{\text{LO}} \mathcal{R}_{\text{LO}}^\lambda \times 4\alpha_s C_F \frac{e^{-i(\mathbf{k}_{g\perp} - \frac{z_g}{z_1} \mathbf{k}_{1\perp}) \cdot \mathbf{r}_{xx'}}}{(\mathbf{k}_{g\perp} - \frac{z_g}{z_1} \mathbf{k}_{1\perp})^2},$$

and integrate it over the out-of-cone gluon phase space. With $\xi = z_g/z_1$ and the transverse integral cut by the jet condition [CSSV (B.9)],

$$\int \frac{d^2 \mathcal{C}_{gq\perp}}{(2\pi)^2} \frac{e^{-i\mathcal{C}_{gq\perp} \cdot \Delta}}{c_{gq\perp}^2} \Theta(c_{gq\perp}^2 - R^2 \mathbf{k}_{1\perp}^2 \xi^2) = -\frac{1}{4\pi} \ln \left(\frac{\mathbf{k}_{1\perp}^2 \Delta^2 R^2 \xi^2}{c_0^2} \right) + \mathcal{O}(R^2), \quad c_0 = 2e^{-\gamma_E},$$

one obtains [CSSV (B.7)]

$$d\sigma_{R_2 \times R_2, \text{out, soft-div}} = \dots \times \Xi_{\text{LO}} \frac{\alpha_s C_F}{\pi} \left\{ \ln^2 \left(\frac{z_1}{z_0} \right) - \ln \left(\frac{z_1}{z_0} \right) \ln \left(\frac{\mathbf{k}_{1\perp}^2 \mathbf{r}_{xx'}^2 R^2}{c_0^2} \right) \right\} + \mathcal{O}(R^2).$$

Read the structure. The $\ln^2(z_0)$ is (*rapidity log*) $\times$ (*angular log*): the ξ^2 inside the transverse logarithm is angular ordering, and $\int_{z_0/z_1}^1 \frac{d\xi}{\xi} 2 \ln \xi = -\ln^2(z_0/z_1)$. This double log is **not** part of the JIMWLK kernel — leading-log rapidity evolution is single-logarithmic. It cancels, at the level of the IRC-safe dijet observable, against the in-cone real contribution and the virtual V_3 diagram. If a $\ln^2(1/z_0)$ survives in your impact factor, you have made an error.

The corresponding statements for the other colour channel: the interference $R_2 \times R_2'$ soft-divergent piece is CSSV (B.17), and integrated, CSSV (B.20),

$$d\sigma_{R_2 \times R_2', \text{soft-div}} = \dots \times \frac{\alpha_s}{\pi} \int_{z_0}^{z_1} \frac{dz_g}{z_g} \left[2 \ln \left(\frac{z_g}{z_1 z_2} \right) + \ln \left(\frac{\mathbf{P}_\perp^2 \mathbf{r}_{xy'}^2}{c_0^2} \right) \right] \Xi_{\text{NLO},3},$$

which again produces $\ln^2 z_0$, cancelling against the $z_g \rightarrow 0$ singularity of $V_3 \times \text{LO}$ [CSSV (B.30)].

2.3 (D3) Final-state collinear divergence

Limit: $\mathbf{z}_\perp \rightarrow \mathbf{x}_\perp$ (or $\mathbf{y}_\perp$) at fixed z_g ; in momentum space $\mathbf{k}_{g\perp}/z_g \rightarrow \mathbf{k}_{1\perp}/z_1$. **Regulator:** dimensional regularization in the transverse plane, $d = 2 \rightarrow d = 2 - \varepsilon$ (CSSV's convention; other papers use $2 - 2\varepsilon$). **Strength:** single $1/\varepsilon$.

Because $\int d^2 \mathbf{z}_\perp / \mathbf{r}_{zx}^2$ is logarithmically divergent at short distance, the real trijet has a $1/\varepsilon$ pole whenever the gluon becomes collinear to the quark or the antiquark. By Kinoshita–Lee–Nauenberg this cancels once the final state is defined inclusively over degenerate configurations — i.e. once a **jet algorithm** is applied. Explicitly, the virtual pole [CSSV (B.23)]

$$d\sigma_{\text{IR} \times \text{LO}} \propto \Xi_{\text{LO}} \frac{\alpha_s C_F}{2\pi} \left\{ \left(\ln \frac{z_1}{z_0} + \ln \frac{z_2}{z_0} - \frac{3}{2} \right) \left(\frac{2}{\varepsilon} + \ln(e^{\gamma_E} \pi \mu^2 \mathbf{r}_{xy}^2) \right) + \frac{1}{2} \ln^2 \frac{z_2}{z_1} - \frac{\pi^2}{6} + \frac{5}{2} - \frac{1}{2} \right\}$$

is cancelled by the in-cone real emission [CSSV (B.24)]

$$d\sigma_{R_2 \times R_2, \text{in}} = d\sigma_{\text{LO}} \frac{\alpha_s C_F}{\pi} \left\{ \left(\frac{3}{4} - \ln \frac{z_1}{z_0} \right) \frac{2}{\varepsilon} + \ln^2 z_1 - \ln^2 z_0 - \frac{\pi^2}{6} + \left(\ln \frac{z_1}{z_0} - \frac{3}{4} \right) \ln \left(\frac{R^2 \mathbf{k}_{1\perp}^2}{\tilde{\mu}^2 z_1^2} \right) + \frac{1}{4} + \frac{3}{2} \left(1 - \ln \frac{z_1}{2} \right) \right\}$$

plus its $q \leftrightarrow \bar{q}$ mirror. The physical remnant is the jet-radius logarithm. After the dust settles the coefficient of $\ln R$ in the finite dijet weight is $-3 \ln R = -\frac{3}{2} \ln R^2$ [CSSV (3.14)], which is precisely the finite part of the z -integrated DGLAP P_{qq} splitting function — the collinear-factorization fingerprint.

2.4 (D4) Ultraviolet divergence — a virtual-only statement

The transverse UV region $\mathbf{z}_\perp \rightarrow \mathbf{x}_\perp$ with the gluon *reabsorbed* produces UV $1/\varepsilon$ poles in SE_1 , SE_2 , SE_3 and V_2 ; these cancel among themselves and against the in-cone real [CSSV §3.2, footnote 13]. A trijet with three *resolved* jets has no UV divergence. It is listed here only so that the inventory is complete and so that you do not double-count SE_1 , whose UV-divergent part belongs to the $\Xi_{\text{NLO},1}$ structure and whose UV-finite part belongs to $d\sigma_{V, \text{no-sud, other}}$.

2.5 What is not divergent (and why that is a feature of the CGC)

- **No initial-state collinear singularity.** The projectile is a photon; the target is a shockwave, resummed to all orders in $g^2 A^{1/3}$. There is no PDF to factorize into. The role played by μ_f and DGLAP in collinear factorization is played here by z_f and JIMWLK. This is the single biggest structural difference from Harris–Owens.
- **No infrared divergence at large $|\mathbf{z}_\perp|$.** Colour transparency / colour neutrality: the correlators $\Xi_{\text{NLO},i}$ vanish when the emitted gluon is placed far from the $q\bar{q}$ pair, so $\mathbf{k}_{g\perp} \rightarrow 0$ at fixed z_g is regulated by the target itself (physically, by Q_s). Do not introduce a transverse-momentum cutoff for it.
- **No divergence at $z_g \rightarrow 1$ or at $z_1, z_2 \rightarrow 0$ for fixed jet kinematics.**

2.6 Summary table

Label	Limit	Diagrams	Regulator	Strength	Fate
D1 rapidity / slow	$z_g \rightarrow 0, \mathbf{k}_{g\perp}$ fixed	all real + virtual	Λ^-, z_0	$\ln \frac{1}{z_0}$	Absorbed into B-JIMWLK / BK. This is piece (I).
D2 soft	$k_g^\mu \rightarrow 0$	R_2, R'_2 only (gluon after SW)	Λ^-	$\ln^2 \frac{1}{z_0}$	Cancels against in-cone real + V_3 . Leaves the Sudakov logs.
D3 final-state collinear	$\mathbf{z}_\perp \rightarrow \mathbf{x}_\perp, \mathbf{y}_\perp$	R_2, R'_2 only	$d = 2 - \epsilon$	$1/\epsilon$	Cancels (KLN) against virtual IR pole once a jet algorithm is applied. Leaves $\ln R$.
D4 UV	$\mathbf{z}_\perp \rightarrow \mathbf{x}_\perp$, gluon reabsorbed	$SE_{1,2,3}, V_2$ (virtual)	$d = 2 - \epsilon$	$1/\epsilon$	Cancels internally among virtual diagrams + in-cone real.
—	$ \mathbf{z}_\perp \rightarrow \infty$	gluon crossing SW	none needed	finite	Regulated by colour neutrality of the target.
—	initial state	—	—	absent	No PDF factorization in DIS off a shockwave.

3. Two-cutoff phase-space slicing, translated to the CGC

3.1 The method in its original setting

Harris–Owens organize an NLO calculation with a $2 \rightarrow 2$ Born, $2 \rightarrow 3$ real emission and one-loop virtual corrections by **slicing the real three-body phase space**, hierarchically, with two small dimensionless cutoffs:

$$\sigma = \underbrace{\frac{1}{2\Phi} \int_S \sum |M_3|^2 d\Gamma_3}_{\text{soft}} + \underbrace{\frac{1}{2\Phi} \int_{\text{HC}} \sum |M_3|^2 d\Gamma_3}_{\text{hard-collinear}} + \underbrace{\frac{1}{2\Phi} \int_{\text{HC}} \sum |M_3|^2 d\Gamma_3}_{\text{hard, non-collinear}} \quad \text{HO (2.1),(2.2),(2.24)}$$

- **S:** $0 \leq E_5 \leq \delta_s \sqrt{s_{12}}/2$ (gluon 5 soft, in the partonic CM frame). In S the matrix element is replaced by the **eikonal (double-pole) approximation**, $|M_3|^2|_{\text{soft}} \simeq -g^2 \mu_r^{2\epsilon} \sum_{f,f'} \frac{p_f \cdot p_{f'}}{(p_f \cdot p_5)(p_{f'} \cdot p_5)} M_{ff'}^0$ [HO (2.20)], and the gluon is integrated analytically over *all* of its angular phase space in $n = 4 - 2\epsilon$ dimensions. Result: a **two-body** weight containing $1/\epsilon^2$, $1/\epsilon$, $\ln^2 \delta_s$, $\ln \delta_s$.
- **HC:** $E_5 > \delta_s \sqrt{s_{12}}/2$ but some invariant $s_{ij} \leq \delta_c s_{12}$. Here the matrix element is replaced by the **leading-pole (splitting-function) approximation**, $\sum |M_3|^2 \simeq \sum |M_2|^2 P_{ij}(z, \epsilon) g^2 \mu_r^{2\epsilon} \frac{2}{s_{45}}$ [HO (2.32)], and integrated over the collinear slice. Result: a second **two-body** weight, with $1/\epsilon$, $\ln \delta_c$, $\ln \delta_s$ and the DGLAP kernels.
- **HC:** everything else. **Finite**, computed numerically in exactly four dimensions with a Monte Carlo, keeping full three-body kinematics.

Adding the virtual corrections, all ϵ poles cancel [HO (3.109)–(3.110): $A_2^v + A_2^s = 0$, $A_1^v + A_1^s + A_1^{\text{coll}} = 0$], and the answer is two finite pieces — a two-body weight depending explicitly on $\ln \delta_s$, $\ln^2 \delta_s$, $\ln \delta_c$ and a three-body weight depending on them through the boundary of its integration region. When histogrammed together for an infrared-safe observable, all cutoff dependence cancels, provided $\delta_c \ll \delta_s \ll 1$.

Why slicing and not subtraction, here. Subtraction (dipole, FKS) needs a local counterterm that reproduces the full singular behaviour of the matrix element point-by-point in phase space. In the CGC the "matrix element" is a multi-dimensional transverse-coordinate integral of products of Wilson-line correlators, and a local counterterm would have to be constructed in the eight-dimensional $\mathbf{X}_\perp$ space simultaneously with the z_g integration. Slicing needs only the *integrated* approximants in the singular regions, which are exactly the objects one already computes analytically (the eikonal limit, the collinear limit, the slow limit). That is why every existing small- x NLO dijet calculation is, in effect, a slicing calculation — including CSSV, whose "in-cone / out-of-cone" and "soft-div / sud2 / no-Sud" split is a two-cutoff slice, written in a different language.

3.2 The dictionary

Harris–Owens (collinear factorization)	This problem (CGC, DIS at small x)
Born $2 \rightarrow 2$; real $2 \rightarrow 3$; one-loop $2 \rightarrow 2$	$\gamma^* \rightarrow q\bar{q}$ (dijet); $\gamma^* \rightarrow q\bar{q}g$ (trijet); one-loop $q\bar{q}$
Soft cutoff δ_s on the gluon energy, $E_5 \leq \delta_s \sqrt{s_{12}}/2$	Soft cutoff δ on the gluon <i>minus-momentum fraction</i> , $z_g \leq \delta$. (In the dipole frame z_g , not $ \mathbf{k}_{g\perp} $, is the soft variable: the dz_g/z_g of the measure is the only unregulated integration.)
Collinear cutoff δ_c on invariants, $s_{ij} \leq \delta_c s_{12}$	Jet radius R : in-cone if $\mathcal{C}_{ik\perp}^2 \leq R^2 \mathbf{p}_j^2 \min(z_i^2, z_k^2)/z_j^2$ [CSSV (3.4)], with $\mathcal{C}_{ik\perp} = \frac{z_i}{z_j} \mathbf{p}_{k\perp} - \frac{z_k}{z_j} \mathbf{p}_{i\perp}$. Unlike δ_c , R is <i>physical</i> and is not taken to zero.
Eikonal / double-pole approximation in S	The $z_g \rightarrow 0$ limit of the $q\bar{q}g$ light-cone wave function; produces the JIMWLK dipole kernels and, from $\mathcal{M}_{\text{aft}}$, the eikonal factor $1/(\mathbf{k}_{g\perp} - \frac{z_g}{z_1} \mathbf{k}_{1\perp})^2$.
Leading-pole approximation in HC, $P_{ij}(z, \epsilon)$	Same DGLAP kernels P_{qq}, P_{gq} : the collinear limit of the trijet wave function factorizes into $P_{qq} \otimes (\text{LO dipole})$. Fingerprint in the answer: $-\frac{3}{2} \ln R^2$.
Initial-state collinear pole absorbed into the PDF at μ_f ($\overline{\text{MS}}$)	Rapidity divergence $\ln(1/z_0)$ absorbed into the CGC weight functional $W_{Y_f}[\rho_A]$ at $Y_f = \ln(1/z_f)$.
μ_f -independence $\Leftrightarrow$ DGLAP: $\partial_{\ln \mu_f^2} G = P \otimes G$	z_f -independence $\Leftrightarrow$ B-JIMWLK: $\partial_{Y_f} W_{Y_f}^{\text{LL}} = \mathcal{H}_{\text{LL}} W_{Y_f}^{\text{LL}}$ [CSSV (3.35)]; at large N_c and for the dipole, BK.
Two-body weight carries $\ln \delta_s, \ln^2 \delta_s, \ln \delta_c$; three-body weight carries the complementary dependence; sum is cutoff-independent	Identical statement with $\delta_s \rightarrow \delta, \delta_c \rightarrow R$ — except that R -dependence does <i>not</i> cancel, because R defines the observable. Only the δ -dependence cancels.
Validity: $\delta_c \ll \delta_s \ll 1$, drop $\mathcal{O}(\delta_c), \mathcal{O}(\delta_s)$	Validity: $z_0 \ll z_f \lesssim \delta \ll 1$ and $R^2 \ll 1$; drop $\mathcal{O}(\delta)$ and $\mathcal{O}(R^2)$. Additionally $\alpha_s \ln(z_{1,2}/z_f) \ll 1$.

3.3 The genuinely new feature: the slice is two-dimensional

In Harris–Owens the factorization scale μ_f enters through a one-dimensional renormalization group: you cut the collinear region at δ_c , subtract the $\overline{\text{MS}}$ counterterm, and what remains depends on $\ln \mu_f$ linearly. Nothing about the soft region interferes with that.

Here it does. The rapidity divergence (D1) and the soft divergence (D2) live in *overlapping* corners of the gluon phase space: D1 is $z_g \rightarrow 0$ at any $\mathbf{k}_{g\perp}$, D2 is $z_g \rightarrow 0$ with $\mathbf{k}_{g\perp} \rightarrow 0$. Consequently:

A cut of the form " $z_g < z_f$, all $\mathbf{k}_{g\perp}$ " is the wrong slicing surface. It assigns to the evolution part of the phase space where the soft (Sudakov) approximation, not the slow (eikonal-in- z_g) approximation, is the valid one. The correct surface is fixed by **lifetime ordering** and is a wedge, not a half-plane. This is §5, and it is the reason the second uploaded paper is needed.

4. Slicing the trijet, region by region

4.1 The regions

Let $\mathcal{M}_2 / \mathcal{M}_3$ be the (IRC-safe) measurement functions selecting a two-jet / three-jet configuration. Define, hierarchically:

$$\begin{aligned}
\mathbf{S} &= \{z_0 < z_g < \delta\}, \text{ all } \mathbf{k}_{g\perp}. \\
\mathbf{H} &= \{z_g > \delta\}, \text{ and within it} \\
\mathbf{HC} &= \mathbf{H} \cap \{\text{gluon in-cone with } q \text{ or } \bar{q}\} = \mathbf{H} \cap \{\mathcal{C}_{gq\perp}^2 \leq R^2 \mathbf{k}_{1\perp}^2 \min(1, z_g^2/z_1^2) \text{ or } (q \rightarrow \bar{q})\}, \\
\overline{\mathbf{HC}} &= \mathbf{H} \setminus \mathbf{HC} \text{ — three resolved jets.}
\end{aligned}$$

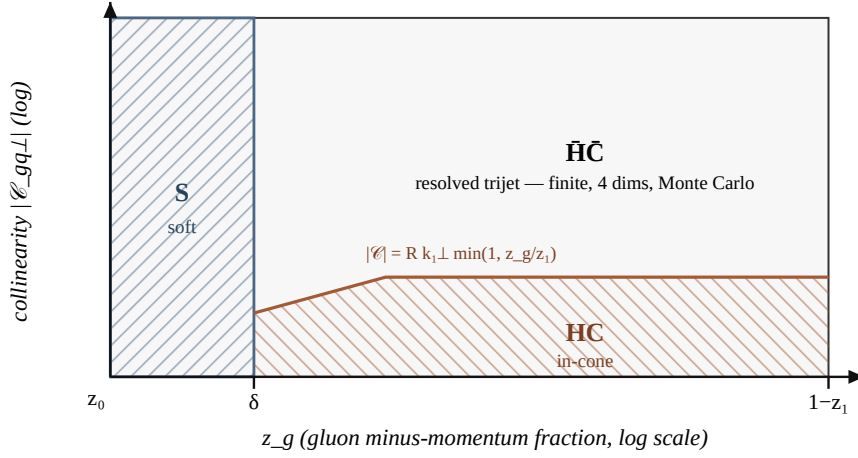


Figure 1. Hierarchical slicing of the gluon phase space. **S** (soft, $z_g < \delta$): eikonal approximation, integrated analytically over the whole transverse plane in $d = 2 - \varepsilon$; two-jet kinematics (k_1, k_2) . **HC** (hard, in-cone): collinear approximation, integrated over the cone in $d = 2 - \varepsilon$; two-jet kinematics $(k_1 + k_g, k_2)$ or $(k_1, k_2 + k_g)$. **$\overline{\text{HC}}$** : exact trijet, finite, four dimensions. The soft strip is defined *first* and spans all collinearity, so S and HC never overlap; this is the Harris–Owens hierarchy, HO (2.1)–(2.2).

Then, identically,

$$\int_{\Phi_g} d\sigma_3^{\text{LO}} \mathcal{M} = \underbrace{W_S(\delta, \varepsilon, z_0)}_{\text{2-body, analytic}} + \underbrace{W_{\text{HC}}(\delta, R, \varepsilon)}_{\text{2-body, analytic}} + \underbrace{\int_{\overline{\text{HC}}} d\sigma_3^{\text{LO}} \mathcal{M}_3}_{\text{3-body, finite}} + \mathcal{O}(\delta, R^2).$$

4.2 Region S — the soft weight

In S the gluon is unresolved: up to $\mathcal{O}(\delta)$ it does not change the jet momenta, so the measurement function is the LO one, $\mathcal{M} \rightarrow \mathcal{M}_2(k_1, k_2)$, whether the gluon happens to lie inside or outside a cone. Replace the trijet by its $z_g \rightarrow 0$ limit and integrate over *all* $\mathbf{k}_{g\perp}$ in $d = 2 - \varepsilon$:

$$W_S = \int_{z_0}^{\delta} \frac{dz_g}{z_g} \int \frac{d^{2-\varepsilon} \mathbf{k}_{g\perp}}{(2\pi)^{2-\varepsilon}} \left[d\sigma_3^{\text{LO}} \right]_{\text{eik}} \mathcal{M}_2.$$

Two structurally different things come out of this single integral, and they must be separated by hand:

1. **The single-log (D1) piece**, from the whole $\mathbf{k}_{g\perp}$ range with the z_g -independent dipole kernels — this is $\mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}} \ln(\delta/z_0)$.
2. **The double-log (D2) piece and the $1/\varepsilon$ collinear pole (D3)**, from the region where the transverse integral is itself logarithmic with a z_g -dependent boundary.

For the Ξ_{LO} colour channel the out-of-cone part of this integral is CSSV (B.7), quoted in §2.2; for the $\Xi_{\text{NLO},3}$ channel it is CSSV (B.20). Both carry $\ln^2 z_0$, and both carry $\ln z_0$.

4.3 Region HC — the in-cone collinear weight

Here $z_g > \delta$ but the gluon is clustered with the quark (or antiquark) into a single jet of momentum $p_J^\mu = k_1^\mu + k_g^\mu$, $z_J = z_1 + z_g$. The trijet matrix element is replaced by its leading collinear behaviour,

$$d\sigma_3^{\text{LO}} \Big|_{\text{coll}} \simeq d\sigma_{\text{LO}}(p_J, k_2) \otimes \frac{\alpha_s}{2\pi} \frac{P_{qq}(z, \varepsilon)}{C_{gq\perp}^2}, \quad P_{qq}(z, \varepsilon) = C_F \frac{1+z^2}{1-z} + \varepsilon P'_{qq}(z), \quad P'_{qq} = -C_F(1-z),$$

[HO (2.32),(2.37)–(2.38)], with $z = z_1/z_J$, and integrated over the cone $\mathcal{C}_{gq\perp}^2 \leq R^2 \mathbf{p}_J^2 \min(z^2, (1-z)^2)$ in $d = 2 - \varepsilon$. The result is CSSV (B.24) (plus its $q \leftrightarrow \bar{q}$ mirror), quoted in §2.3. It contains:

$$W_{\text{HC}} \supset \frac{2}{\varepsilon} \left(\frac{3}{4} - \ln \frac{z_1}{z_0} \right), \quad \ln^2 z_1 - \ln^2 z_0, \quad \left(\ln \frac{z_1}{z_0} - \frac{3}{4} \right) \ln \frac{R^2 \mathbf{k}_{1\perp}^2}{\tilde{\mu}^2 z_1^2},$$

with $\tilde{\mu}^2 = 4\pi e^{-\gamma_E} \mu^2$ the $\overline{\text{MS}}$ scale. Note the $3/4$: that is $\frac{3}{4} = \frac{1}{2} \int_0^1 dz \frac{1+z^2}{1-z} \Big|_{\text{reg}}$, the finite part of the z -integrated P_{qq} , and it is what becomes the $-3 \ln R$ of the final answer.

4.4 Region $\overline{HC}$ — the finite trijet you integrate numerically

No approximation: the exact $q\bar{q}g$ cross-section, in exactly four dimensions, with the constraints $z_g > \delta$ and both out-of-cone conditions imposed. This is the object CSSV call the "no-Sud" real cross-section: Eqs. (3.21), (3.22) (gluon *not* crossing the shockwave) and (3.23) (gluon crossing). CSSV state explicitly that this piece "is free of any divergence, as well as large back-to-back logarithms, for all kinematic regions of the final state" — which is exactly the property you want from piece (IV).

Two practical remarks:

- In (3.23) the $z_g \rightarrow 0$ limit has already been subtracted by the term labelled "slow" (the last line of that equation), with a $\Theta(z_f - z_g)$. That subtraction is *the same object* as the one you are performing here; do not do it twice.
- The upper limit of the z_g integral in the analytic weights is arbitrary (momentum conservation gives $z_g \leq 1 - z_1$, but any choice differs by finite terms which are compensated in $\overline{HC}$). CSSV take z_1 for $R_2 \times R_2$ and z_2 for $R'_2 \times R'_2$ to preserve $q \leftrightarrow \bar{q}$ symmetry [discussion after CSSV (B.6) and (B.21)].

4.5 The virtual dijet

The one-loop $q\bar{q}$ contribution supplies (i) the UV poles that cancel among $SE_{1,2,3}$, V_2 ; (ii) the IR $1/\varepsilon$ pole, CSSV (B.23), that cancels against W_{HC} ; (iii) its own dz_g/z_g rapidity divergence — the V_1, V_3, SE_1 diagrams have a loop z_g integral running down to zero — which supplies the *virtual* half of the JIMWLK kernel; and (iv) a $\ln^2 z_0$ from V_3 , CSSV (B.30), that cancels against (B.20).

4.6 Assembling: the two cancellations

(a) ε -poles and $\ln^2 z_0$ cancel. For the Ξ_{LO} channel [CSSV (B.25)]:

$$\begin{aligned} & d\sigma_{IR \times LO} + d\sigma_{LO \times IR} + d\sigma_{R_2 \times R_2, in} + d\sigma_{R_2 \times R_2, out, soft-div} + (R_2 \rightarrow R'_2) \\ &= \dots \Xi_{LO} \frac{\alpha_s C_F}{\pi} \left\{ -\ln \frac{z_1}{z_0} \ln \frac{\mathbf{r}_{xx'}^2}{|\mathbf{r}_{xy}| |\mathbf{r}_{x'y'}|} - \ln \frac{z_2}{z_0} \ln \frac{\mathbf{r}_{yy'}^2}{|\mathbf{r}_{xy}| |\mathbf{r}_{x'y'}|} - \frac{3}{4} \ln \frac{\mathbf{k}_{1\perp}^2 \mathbf{k}_{2\perp}^2 \mathbf{r}_{xy}^2 \mathbf{r}_{x'y'}^2}{c_0^4} \right. \\ &\quad \left. - 3 \ln R + \frac{1}{2} \ln^2 \frac{z_1}{z_2} + \frac{11}{2} + 3 \ln 2 - \frac{\pi^2}{2} \right\} + \mathcal{O}(R^2). \end{aligned}$$

No $1/\varepsilon$; no $\ln^2 z_0$. Only **single** $\ln(1/z_0)$ survives — which is the statement that rapidity factorization holds at leading logarithmic accuracy. For the $\Xi_{NLO,3}$ channel the same happens, CSSV (B.31).

(b) Isolating the surviving single log. Introduce $z_f = k_f^-/q^-$ and split $\ln(1/z_0) = \ln(z_f/z_0) + \ln(1/z_f)$. The $\ln(z_f/z_0)$ part is [CSSV (B.26), (B.32)]

$$\begin{aligned} d\sigma_{LO,slow} &= \dots \frac{\alpha_s C_F}{\pi} \ln \left(\frac{k_f^-}{\Lambda^-} \right) \ln \left(\frac{\mathbf{r}_{xy}^2 \mathbf{r}_{x'y'}^2}{\mathbf{r}_{xx'}^2 \mathbf{r}_{yy'}^2} \right) \Xi_{LO}, \\ d\sigma_{NLO,3,slow} &= \dots \frac{\alpha_s}{\pi} \ln \left(\frac{k_f^-}{\Lambda^-} \right) \ln \left(\frac{\mathbf{r}_{xy}^2 \mathbf{r}_{y'x'}^2}{\mathbf{r}_{xy}^2 \mathbf{r}_{x'y'}^2} \right) \Xi_{NLO,3}, \end{aligned}$$

and together with the analogous $z_\perp$ -dependent channels they reconstruct CSSV (3.29) $= \ln(k_f^-/\Lambda^-) \mathcal{H}_{LL} \otimes d\sigma_{LO}$. The $\ln(1/z_f)$ part stays behind as a finite, z_f -dependent contribution to the impact factor — CSSV (B.27), (B.33), (B.34).

At this point you have achieved what you asked for, *except* that the split in (b) was made with a naive $z_g < z_f$ cut. Correcting that is §5.

5. The rapidity divergence done properly: lifetime ordering

5.1 Where the naive slow limit breaks down

The naive prescription is: in the term $\ln(k_f^-/\Lambda^-)\mathcal{H}_{LL} \otimes d\sigma_{LO}$, "set $z_g = 0$ everywhere except in the $1/z_g$ divergence" [CSSV, above (4.39)]. Look at where z_g actually sits. The trijet wave functions contain

$$K_0(QX_R), \quad K_0(QX_V), \quad X_R^2 = z_1 z_2 \mathbf{r}_{xy}^2 + z_1 z_g \mathbf{r}_{zx}^2 + z_2 z_g \mathbf{r}_{zy}^2, \\ X_V^2 = z_2(1 - z_g) \mathbf{r}_{xy}^2 + z_g(z_1 - z_g) \mathbf{r}_{zx}^2 + z_2 z_g \mathbf{r}_{zy}^2 \quad [\text{CSSV (4.40),(4.41)}].$$

Setting $z_g = 0$ in $X_{R/V}$ recovers the LO wave function $K_0(\bar{Q}r_{xy})$. But $\mathbf{z}_\perp$ is *integrated*: no matter how small z_g is, there is always a region of large $|\mathbf{r}_{zx}|, |\mathbf{r}_{zy}|$ in which the transverse size compensates the smallness of z_g and the z_g -suppressed terms are *not* negligible. Setting $z_g = 0$ there is simply wrong, and the phase space you assign to the evolution is too large.

In back-to-back kinematics the two effective sizes simplify to a common form [CSSV (4.42)],

$$X_R^2 \approx X_V^2 \approx z_1 z_2 \mathbf{u}_\perp^2 + z_g \mathbf{r}_{zb}^2,$$

with $\mathbf{u}_\perp = \mathbf{x}_\perp - \mathbf{y}_\perp$ and $\mathbf{b}_\perp = z_1 \mathbf{x}_\perp + z_2 \mathbf{y}_\perp$. The approximation $K_0(QX_{R/V}) \rightarrow K_0(\bar{Q}u_\perp)$ is therefore legitimate **only** when

$$z_g \mathbf{r}_{zb}^2 \leq z_1 z_2 \mathbf{u}_\perp^2 \quad [\text{CSSV (4.43)}]$$

5.2 The constraint is lifetime ordering

Read the constraint in momentum space. Parametrically $\mathbf{u}_\perp^2 \sim 1/Q^2$ and $\mathbf{r}_{zb}^2 \sim 1/\mathbf{k}_{g\perp}^2$, so (4.43) becomes [CSSV (4.44)]

$$\frac{z_g}{\mathbf{k}_{g\perp}^2} \leq \frac{z_1 z_2}{Q^2} \quad \Longleftrightarrow \quad \frac{1}{k_g^+} \leq z_1 z_2 \frac{2q^-}{Q^2} \sim \frac{1}{q^+}, \quad k_g^+ = \frac{\mathbf{k}_{g\perp}^2}{2z_g q^-}.$$

Since $1/k^+$ is the lifetime (coherence time) of a light-cone fluctuation of plus-momentum k^+ , this says:

the emitted gluon fluctuation must be shorter-lived than the parent $q\bar{q}$ dipole:

$$\tau_g = \frac{1}{k_g^+} \leq \frac{1}{k_f^+} = \tau_{q\bar{q}}, \quad 2k_f^+ k_f^- \equiv Q_f^2, \quad Q_f^2 \sim Q^2 \sim \mathbf{P}_\perp^2.$$

This is the small- x analogue of angular ordering / the BFKL "consistency condition": successive emissions in the projectile cascade are ordered *both* in k^- (decreasing, the usual rapidity ordering $k_g^- \leq k_f^-$) *and* in k^+ (increasing, i.e. decreasing lifetime, $k_g^+ \geq k_f^+$).

In coordinate space, using $\mathbf{k}_{g\perp}^2 \sim 1/\mathbf{r}_{zb}^2 \sim 1/\mathbf{r}_{zb'}^2$ and symmetrizing between amplitude and conjugate amplitude, the condition $1/k_g^+ \leq 1/k_f^+$ reads [CSSV (4.47)]

$$k_g^- \min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) \leq \frac{k_f^-}{Q_f^2}, \quad \text{equivalently} \quad \frac{z_g}{z_f} \min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2 \leq 1.$$

5.3 The correct slicing surface: a wedge, not a half-plane

Introduce the two logarithmic variables that span the emission ("Lund") plane of the slow gluon:

$$t \equiv \ln \frac{z_f}{z_g} \quad (\geq 0 \text{ for a slow gluon}), \quad \rho \equiv \ln(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2) \simeq \ln \frac{Q_f^2}{\mathbf{k}_{g\perp}^2}.$$

Then:

- rapidity ordering $z_g < z_f \iff t > 0$;
- lifetime ordering $k_g^+ \geq k_f^+ \iff \rho < t$.

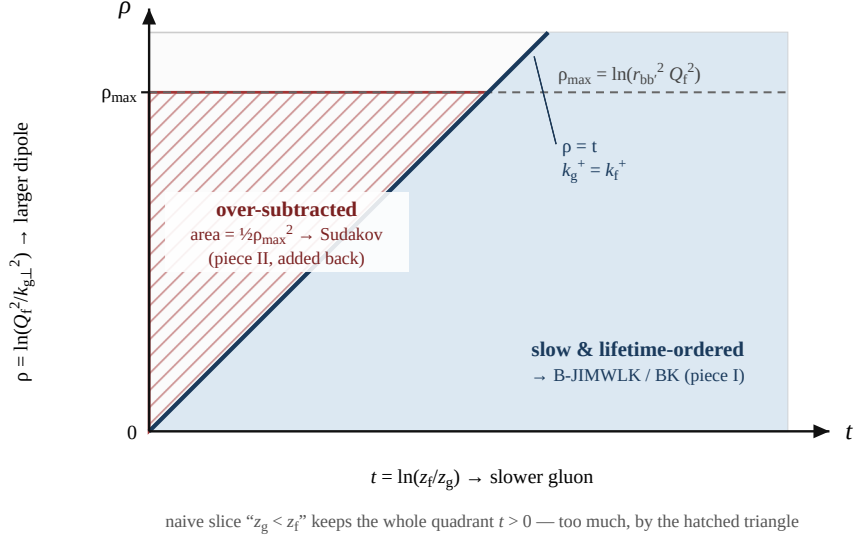


Figure 2. The emission plane of the slow gluon. The naive rapidity slice keeps the entire quadrant $t > 0$. Lifetime ordering restricts the evolution phase space to the wedge $\rho < t$ below the diagonal. The hatched triangle $0 < t < \rho < \rho_{\max}$ is what a naive slice wrongly hands to the evolution; its area, $\frac{1}{2}\rho_{\max}^2$ with $\rho_{\max} = \ln(r_{bb'}^2 Q_f^2)$, is the Sudakov double logarithm that must be added back to the impact factor with a *negative* sign.

5.4 The over-subtraction, computed

Quantitatively: the unconstrained ("naive") leading-log rapidity evolution of the Weizsäcker–Williams gluon TMD, obtained as the back-to-back/correlation limit of CSSV (3.29), is [CSSV (4.45)]

$$\hat{G}_{Y_f}^{ij}(\mathbf{b}_\perp, \mathbf{b}'_\perp) = \hat{G}_{Y_0}^{ij} + \frac{\alpha_s N_c}{2\pi^2} \int_{Y_0}^{Y_f} dY \int d^2 \mathbf{z}_\perp \left\{ -\frac{\mathbf{r}_{bb'}^2}{\mathbf{r}_{zb}^2 \mathbf{r}_{zb'}^2} \hat{G}_Y^{ij} + \text{other correlators} \right\},$$

with $\hat{G}_Y^{ij}(\mathbf{b}_\perp, \mathbf{b}'_\perp)$ the coordinate-space WW distribution, CSSV (2.22)–(2.23). Imposing the lifetime constraint of §5.2 gives the **kinematically improved** equation [CSSV (4.48)]

$$\hat{G}_{Y_f}^{ij} = \hat{G}_{Y_0}^{ij} - \frac{\alpha_s N_c}{2\pi^2} \int_{Y_0}^{Y_f} dY \int d^2 \mathbf{z}_\perp \Theta\left(Y_f - Y - \ln\left(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2\right)\right) \frac{\mathbf{r}_{bb'}^2}{\mathbf{r}_{zb}^2 \mathbf{r}_{zb'}^2} \hat{G}_Y^{ij} + \dots$$

Using this instead of the unconstrained equation amounts to **subtracting** from the evolution, and therefore **adding back** to the impact factor, exactly the hatched triangle of Figure 2 [CSSV (4.50)]:

$$\Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi^2} \hat{G}_{Y_f}^{ij}(\mathbf{b}_\perp, \mathbf{b}'_\perp) \int_{z_0}^{z_f} \frac{dz_g}{z_g} \int d^2 \mathbf{z}_\perp \Theta\left(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2 - \frac{z_f}{z_g}\right) \frac{\mathbf{r}_{bb'}^2}{\mathbf{r}_{zb}^2 \mathbf{r}_{zb'}^2}.$$

This integral is elementary (it is done in full in **Appendix A**); the result is convergent as $z_0 \rightarrow 0$ — the Θ function itself cuts the z_g integral — and equals [CSSV (4.51)]

$$\Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi} \ln^2(Q_f^2 \mathbf{r}_{bb'}^2) \hat{G}_{Y_f}^{ij}(\mathbf{b}_\perp, \mathbf{b}'_\perp) + \mathcal{O}\left(\frac{1}{\mathbf{r}_{bb'}^2 Q_f^2}\right)$$

and, split into its Sudakov double-log, single-log and constant pieces [CSSV (4.52)], with $c_0 = 2e^{-\gamma_E}$:

$$\Delta_{\text{lt}} = \underbrace{-\frac{\alpha_s N_c}{2\pi} \ln^2 \left(\frac{\mathbf{P}_\perp^2 \mathbf{r}_{bb'}^2}{c_0^2} \right) \hat{G}_{Y_f}^{ij}}_{\text{Sudakov double log}} \underbrace{-\frac{\alpha_s N_c}{\pi} \ln \left(\frac{Q_f^2 c_0^2}{\mathbf{P}_\perp^2} \right) \ln \left(\frac{\mathbf{P}_\perp^2 \mathbf{r}_{bb'}^2}{c_0^2} \right) \hat{G}_{Y_f}^{ij}}_{\text{Sudakov single log}} \underbrace{-\frac{\alpha_s N_c}{2\pi} \ln^2 \left(\frac{Q_f^2 c_0^2}{\mathbf{P}_\perp^2} \right)}_{\text{finite}}.$$

5.5 The sign of the double logarithm is your check

Use this as a hard test of your extraction. Soft gluon radiation must *suppress* the back-to-back peak: emitted soft gluons contribute $\mathbf{k}_{g\perp} \sim \mathbf{q}_\perp$ to the imbalance, so the coefficient of $\alpha_s \ln^2(\mathbf{P}_\perp/\mathbf{q}_\perp)$ has to be **negative**.

If you slice with a naive $z_g < z_f$ cut and hand the whole quadrant to JIMWLK, the Sudakov double logarithm you extract from the remaining impact factor comes out **positive** — the wrong sign; see the discussion below CSSV (4.32)–(4.33). Restoring the triangle Δ_{lt} flips it. So: compute the coefficient of $\alpha_s \ln^2$; if it is positive, you have over-subtracted the rapidity divergence.

With Δ_{lt} restored, the complete double + single Sudakov result for a longitudinal photon reads [CSSV (4.53)]

$$\begin{aligned} d\sigma^{(0),\lambda=L} = & \alpha_{\text{em}} \alpha_s e_f^2 \delta_z^{(2)} \frac{8(z_1 z_2)^3 Q^2 \mathbf{P}_\perp^2}{(\mathbf{P}_\perp^2 + Q^2)^4} \int \frac{d^2 \mathbf{r}_{bb'}}{(2\pi)^4} e^{-i\mathbf{q}_\perp \cdot \mathbf{r}_{bb'}} \hat{G}_{Y_f}^0(\mathbf{r}_{bb'}) \left\{ 1 - \frac{\alpha_s N_c}{4\pi} \ln^2 \left(\frac{\mathbf{P}_\perp^2 \mathbf{r}_{bb'}^2}{c_0^2} \right) \right. \\ & \left. - \frac{\alpha_s}{\pi} \left[C_F \ln \left(\frac{1}{z_1 z_2 R^2} \right) - N_c \ln \left(\frac{z_f \mathbf{P}_\perp^2}{z_1 z_2 c_0^2 Q_f^2} \right) \right] \ln \left(\frac{\mathbf{P}_\perp^2 \mathbf{r}_{bb'}^2}{c_0^2} \right) \right\} + \mathcal{O}(\alpha_s), \end{aligned}$$

where the $\hat{G}_{Y_f}^0$ appearing here must be evolved with the **constrained** equation (4.48). The double-log coefficient is now $-\alpha_s N_c/4\pi < 0$, as required.

5.6 The constrained evolution equations you actually feed the divergence into

(i) B-JIMWLK, unconstrained (leading log). The divergence of §2.1 renormalizes the CGC weight functional:

$$\frac{\partial W_{Y_f}^{\text{LL}}[\rho_A]}{\partial Y_f} = \mathcal{H}_{\text{LL}} W_{Y_f}^{\text{LL}}[\rho_A], \quad Y_f = \ln(1/z_f), \quad [\text{CSSV (3.34),(3.35)}]$$

equivalently, for the LO colour structure, $\Xi_{\text{LO}}(\mathbf{X}_\perp|Y_f) = \Xi_{\text{LO}}^{(0)}(\mathbf{X}_\perp|Y_0) + \mathcal{H}_{\text{LL}} \otimes \Xi_{\text{LO}}^{(0)} \ln(z_f/z_0) + \mathcal{O}(\alpha_s^2)$ [CSSV (3.33)]. The initial condition is set at

$$z_0 = \frac{Q_0^2 x_{\text{Bj}}}{Q^2 x_0} \quad [\text{CSSV (3.38)}],$$

which follows from $k_g^+ \leq x_0 P^+$ together with the on-shell relation $2k_g^+ k_g^- = \mathbf{k}_{g\perp}^2$; typically $x_0 = 0.01$ and the MV model is used for W_{Y_0} .

(ii) The whole LO+NLO cross-section then reads [CSSV (3.36),(3.37)]

$$d\sigma_{\text{LO+NLO}} = \int \mathcal{D}[\rho_A] W_{Y_f}^{\text{LL}}[\rho_A] (d\sigma_{\text{LO}} + \alpha_s d\sigma_{\text{NLO,i.f.}}),$$

with the impact factor $d\sigma_{\text{NLO,i.f.}}$ defined as the finite $\mathcal{O}(\alpha_s)$ remainder after the rapidity divergence has been factorized out. Note that the colour correlators inside $d\sigma_{\text{LO}}$ and $d\sigma_{\text{NLO,i.f.}}$ must be defined *without* the $\langle \dots \rangle_Y$ average, which is done once, explicitly, in this formula [CSSV footnote 16].

(iii) Kinematically improved WW evolution — the version you must use if you want the Sudakov sign right — is CSSV (4.48), reproduced in §5.4.

(iv) BK. At large N_c , and for observables controlled by the dipole operator, the JIMWLK hierarchy closes onto the Balitsky–Kovchegov equation for $S_{xy} = \langle D_{xy} \rangle$,

$$\frac{\partial S_{xy}(Y)}{\partial Y} = \frac{\bar{\alpha}_s}{2\pi} \int d^2 \mathbf{z}_\perp \frac{\mathbf{r}_{xy}^2}{\mathbf{r}_{xz}^2 \mathbf{r}_{zy}^2} \left[S_{xz}(Y) S_{zy}(Y) - S_{xy}(Y) \right], \quad \bar{\alpha}_s = \frac{\alpha_s N_c}{\pi},$$

which is just $\mathcal{H}_{\text{LL}}$ acting on a dipole with the quadrupoles factorized. The corresponding kinematic constraint for the dipole operator is [CSSV (4.49)]

$$\Theta \left(Y_f - Y - \ln \left(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{z'b'}^2) / \mathbf{r}_{bb'}^2 \right) \right),$$

i.e. the familiar time-ordering / "kinematically constrained BK" restriction, in which the scale in the logarithm is the **internal** parent-dipole size $1/\mathbf{r}_{bb'}^2$.

5.7 A subtlety worth flagging: WW is *not* BFKL after the improvement

Compare the two constraints:

$$\text{WW TMD: } \Theta(Y_f - Y - \ln(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2)), \quad \text{dipole: } \Theta(Y_f - Y - \ln(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'})/\mathbf{r}_{bb'}^2)).$$

For the WW distribution the constraint involves the **external** kinematic scale $Q_f^2 \sim \mathbf{P}_\perp^2 \sim 1/\mathbf{r}_{xy}^2$ — the hard scale of the *measurement* — and **not** $1/\mathbf{r}_{bb'}^2$. CSSV emphasize the consequence: since the unconstrained rapidity evolution of the WW TMD is BFKL in the dilute limit, after kinematic improvement the dilute limit of the WW evolution is **not** given by the kinematically improved BFKL equation. If you are building a numerical solver, do not recycle a kcBK/kcBFKL kernel for the WW distribution.

The practical reason is easy to see from §5.1: the constraint came from requiring $K_0(QX_{R/V}) \approx K_0(\bar{Q}u_\perp)$, and the scale in that Bessel function is set by the photon virtuality and the jet transverse momentum, i.e. by $\mathbf{u}_\perp$, not by the impact-parameter separation $\mathbf{r}_{bb'}$.

One more remark, since it is easy to worry about it: because k_f^+ is arbitrary, Q_f^2 is arbitrary too, as long as it is parametrically of order $\mathbf{P}_\perp^2$. Different choices shuffle terms between the double log and the single log; the *double*-log coefficient is scheme independent, the *single*-log coefficient is not. CSSV note that there is a choice, $x_f = \frac{Q^2}{c_0^2 \mathbf{P}_\perp^2} \frac{x_1 x_2}{x_{Bj}}$ [CSSV (4.54)], for which the scheme-dependent part of the single-log coefficient vanishes.

6. The final decomposition and a working algorithm

6.1 Master formula

$$d\sigma^{\gamma_A^* + A \rightarrow 2 \text{ jets} + X} = \int \mathcal{D}[\rho_A] W_{Y_f}^{\text{LL, kc}}[\rho_A] \left\{ d\sigma_{\text{LO}} + \alpha_s d\sigma_{\text{NLO, i.f.}} \right\}$$

$$\alpha_s d\sigma_{\text{NLO, i.f.}} = \underbrace{\left[W_S + W_{\text{HC}} + d\sigma_V \right]_{\text{finite}}(\delta, R, z_f)}_{\text{(III) finite dijet weight}} + \underbrace{\Delta_{\text{lt}}(Q_f)}_{\text{(II)}} + \underbrace{\int_{\overline{\text{HC}}(\delta, R)} d\sigma_3^{\text{LO}} \mathcal{M}_3}_{\text{(IV) finite trijet}}$$

$W_{Y_f}^{\text{LL, kc}}$ is the weight functional evolved from $Y_0 = \ln(1/z_0)$ to $Y_f = \ln(1/z_f)$ with the **kinematically constrained** (lifetime-ordered) kernel; that evolution is piece (I). All $1/\varepsilon$, all $\ln^2(1/z_0)$ and all $\ln(1/z_0)$ have been removed.

In the language of CSSV's Eq. (3.6)–(3.7) the same content is written

$$d\sigma_{\text{NLO}} = \left(d\sigma_{R_2 \times R_2, \text{sud2}} + d\sigma_{R_2 \times R_2', \text{sud2}} + R_2 \leftrightarrow R_2' \right) + d\sigma_{\text{sud1}} + d\sigma_{R, \text{no-sud}} + d\sigma_{V, \text{no-sud}} + \ln\left(\frac{k_f^-}{\Lambda^-}\right) \mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}},$$

with the correspondence

This note	CSSV	Content
(I)	$\ln(k_f^-/\Lambda^-) \mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}}$, Eq. (3.29),(3.32)	the entire rapidity divergence
(II)	Eqs. (4.50)–(4.52)	lifetime-ordering add-back; the correct-sign Sudakov double log
(III)	$d\sigma_{\text{sud2}} + d\sigma_{\text{sud1}} + d\sigma_{V, \text{no-sud}}$, Eqs. (3.8),(3.9),(3.12),(3.14),(3.15), (3.17)	finite two-body weight
(IV)	$d\sigma_{R, \text{no-sud}}$, Eqs. (3.21),(3.22),(3.23)	finite resolved trijet

6.2 Algorithm

- Fix the scales.** Choose z_f (e.g. $\min(z_1, z_2)$ or $z_1 z_2$), $Q_f^2 \sim P_\perp^2$ (e.g. $Q_f = P_\perp/c_0$), the jet radius R , and a soft slice δ with $z_0 \ll z_f \lesssim \delta \ll 1$. Require $\alpha_s \ln(z_{1,2}/z_f) \ll 1$.
- Solve the evolution.** Evolve $W_Y[\rho_A]$ (or, in the back-to-back limit, $\hat{G}_Y^{ij}$) from $Y_0 = \ln(1/z_0)$, Eq. (3.38), to Y_f , using the *kinematically constrained* kernel (4.48). Initial condition: MV model at $x_0 \simeq 0.01$. This step consumes the divergence and produces nothing else.
- Trijet Monte Carlo (IV).** Generate $q\bar{q}g$ events with the exact CGC matrix element, impose $z_g > \delta$ and both out-of-cone conditions, cluster into three jets, integrate in four dimensions. Nothing here is divergent.
- Two-body weight (III).** Add the analytic soft weight W_S , the in-cone weight W_{HC} and the virtual $d\sigma_V$; check that $1/\varepsilon$ and $\ln^2 z_0$ cancel, then *subtract* the slow term $\ln(z_f/z_0) \mathcal{H}_{\text{LL}} \otimes d\sigma_{\text{LO}}$ that you already gave to step 2. The explicit results are (B.25) $\rightarrow$ (B.27) and (B.31) $\rightarrow$ (B.33).
- Lifetime add-back (II).** Add $\Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi} \ln^2(Q_f^2 r_{bb'}^2) \hat{G}_{Y_f}^{ij}$. Equivalently and more precisely: the difference between evolving with the constrained kernel (step 2) and the unconstrained one. *Do not do both* — if your evolution code already implements (4.48), Δ_{lt} is **already accounted for** and adding it again double counts.
- Histogram (III) + (IV)** together for your infrared-safe observable.

Step 5 is the one place where it is easy to double count. Piece (II) is *defined* as (unconstrained evolution) – (constrained evolution), moved from the evolution side to the impact-factor side. You either (a) evolve with the naive kernel and add Δ_{lt} to the impact factor, or (b) evolve with the constrained kernel (4.48) and add nothing. Option (b) is preferable because it resums the double transverse logarithms to all orders and stabilizes the evolution beyond leading-log accuracy; option (a) is only correct to $\mathcal{O}(\alpha_s)$.

6.3 Checks to run, in order of usefulness

1. **δ -independence.** Vary δ over a decade (e.g. 10^{-2} to 10^{-3}). (III) and (IV) each move by $\ln \delta, \ln^2 \delta$; their sum must be flat within Monte Carlo errors. This is the standard slicing plateau test (HO §III, Appendix D).
2. **Sign of the Sudakov double log.** Must be negative; see §5.5.
3. **ε -pole cancellation, analytically.** The residues must cancel *before* any numerical work: $A_{1/\varepsilon}^{\text{virt}} + A_{1/\varepsilon}^{\text{in-cone}} = 0$, the analogue of HO (3.109)–(3.110).
4. **Absence of $\ln^2 z_0$.** Independently of the pole cancellation. If it survives, you have mistaken part of the soft region for the slow region.
5. **z_f - and Q_f -independence.** Vary z_f between $z_1 z_2$ and $\min(z_1, z_2)$, and Q_f by a factor of two. The full LO+NLO answer must move only by $\mathcal{O}(\alpha_s^2)$; the size of that motion is your theory-uncertainty band.
6. **$R \rightarrow \text{small}$.** The $\ln R$ dependence of (III) must be $-3 \ln R$ per $q\bar{q}$ pair [CSSV (3.14)] and must be compensated by (IV) up to $\mathcal{O}(R^2)$.
7. **Dilute limit.** Expanding the Wilson lines to lowest order must reproduce the known collinear-factorization NLO $\gamma^* g \rightarrow q\bar{q}$ result for the same observable.

7. Reference formulas for the finite pieces

Collected here so the note is self-contained. All are CSSV results for a *longitudinally* polarized photon; the transverse-photon counterparts are CSSV Appendix C. The overall prefactor $\frac{\alpha_{\text{em}} e_f^2 N_c \delta_z^{(2)}}{(2\pi)^6} \int d^8 \mathbf{X}_\perp e^{-i\mathbf{k}_{1\perp} \cdot \mathbf{r}_{xx'} - i\mathbf{k}_{2\perp} \cdot \mathbf{r}_{yy'}} \mathcal{R}_{\text{LO}}^\lambda$ is abbreviated $[\cdot]$.

(III-a) Sudakov-double-log terms — CSSV (3.8), (3.9)

$$\begin{aligned} d\sigma_{R_2 \times R_2, \text{sud2}} &= [\cdot] C_F \Xi_{\text{LO}} \times \frac{\alpha_s}{\pi} \int_0^1 \frac{d\xi}{\xi} \left[1 - e^{-i\xi \mathbf{k}_{1\perp} \cdot \mathbf{r}_{xx'}} \right] \ln \left(\frac{\mathbf{k}_{1\perp}^2 \mathbf{r}_{xx'}^2 R^2 \xi^2}{c_0^2} \right), \\ d\sigma_{R_2 \times R_2', \text{sud2}} &= [\cdot] \Xi_{\text{NLO},3} \times \frac{-\alpha_s}{\pi} \int_0^1 \frac{d\xi}{\xi} \left[1 - e^{-i\xi \mathbf{k}_{1\perp} \cdot \mathbf{r}_{xy'}} \right] \ln \left(\frac{\mathbf{P}_\perp^2 \mathbf{r}_{xy'}^2 \xi^2}{z_2^2 c_0^2} \right). \end{aligned}$$

The ξ integrals are convergent: the leading rapidity divergence has already been subtracted in the decomposition (B.3),(B.16), so no z_0 appears.

(III-b) Sudakov-single-log terms — CSSV (3.12) = (B.34)

$$\begin{aligned} d\sigma_{\text{sud1}} &= [\cdot] \frac{\alpha_s}{\pi} \left\{ C_F \Xi_{\text{LO}} \left[\ln \left(\frac{z_f}{z_1} \right) \ln \left(\frac{\mathbf{r}_{xx'}^2}{|\mathbf{r}_{xy}| |\mathbf{r}_{x'y'}|} \right) + \ln \left(\frac{z_f}{z_2} \right) \ln \left(\frac{\mathbf{r}_{yy'}^2}{|\mathbf{r}_{xy}| |\mathbf{r}_{x'y'}|} \right) \right] \right. \\ &\quad \left. + \Xi_{\text{NLO},3} \left[\ln \left(\frac{z_1}{z_f} \right) \ln \left(\frac{\mathbf{r}_{xy'}^2}{|\mathbf{r}_{xy}| |\mathbf{r}_{x'y'}|} \right) + \ln \left(\frac{z_2}{z_f} \right) \ln \left(\frac{\mathbf{r}_{yx'}^2}{|\mathbf{r}_{xy}| |\mathbf{r}_{x'y'}|} \right) \right] \right\} \end{aligned}$$

Note the explicit z_f : this term is the residue of the rapidity subtraction scheme, and is where the interplay between slow and soft gluons shows up. It is what makes the single-Sudakov coefficient factorization-scheme dependent.

(III-c) Finite virtual, LO colour structure — CSSV (3.14) = (B.36)

$$d\sigma_{V, \text{no-sud, LO}} = [\cdot] \Xi_{\text{LO}} \frac{\alpha_s C_F}{\pi} \left\{ -\frac{3}{4} \ln \left(\frac{\mathbf{k}_{1\perp}^2 \mathbf{k}_{2\perp}^2 \mathbf{r}_{xy}^2 \mathbf{r}_{x'y'}^2}{c_0^4} \right) - 3 \ln R + \frac{1}{2} \ln^2 \left(\frac{z_1}{z_2} \right) + \frac{11}{2} + 3 \ln 2 - \frac{\pi^2}{2} \right\}$$

The $-3 \ln R = -\frac{3}{2} \ln R^2$ is the finite part of the z -integrated DGLAP P_{qq} — the direct fingerprint of the collinear slice of §4.3. Its resummation for small R would be a DGLAP-like evolution, which CSSV leave aside.

(III-d) Finite virtual, $\Xi_{\text{NLO},3}$ and other structures — CSSV (3.15), (3.17)

$d\sigma_{V, \text{no-sud, NLO},3}$ involves $\bar{Q}_{V3}^2 = z_1 z_2 (1 - z_g/z_1)(1 + z_g/z_2) Q^2$, $\Delta_{V3}^2 = (1 - z_g/z_1)(1 + z_g/z_2) \mathbf{P}_\perp^2$ and the function $\mathcal{J}_\odot(\mathbf{r}_\perp, \mathbf{K}_\perp, \Delta) = \int \frac{d^2 l_\perp}{(2\pi)} \frac{2l_\perp \cdot \mathbf{K}_\perp e^{i l_\perp \cdot \mathbf{r}_\perp}}{l_\perp^2 [(l_\perp - \mathbf{K}_\perp)^2 - \Delta^2 - i\epsilon]}$ [CSSV (3.16)]. Its z_g integral is rendered finite by the explicit subtraction term $K_0(\bar{Q} r_{xy}) \ln(z_g P_\perp r_{xy}/(c_0 z_1 z_2))$ inside (3.15). $d\sigma_{V, \text{no-sud, other}}$, Eq. (3.17), carries the $\Theta(z_f - z_g)$ subtractions that remove its slow divergence; it involves X_V of (3.20).

(IV-a) Finite real, gluon not crossing the shockwave — CSSV (3.21), (3.22)

$$\begin{aligned} d\sigma_{\text{R,no-sud,LO}}^{\gamma_L^*+A\rightarrow q\bar{q}g+X} &= \frac{\alpha_{\text{em}}e_f^2N_c}{(2\pi)^8} \int d^8\mathbf{X}_\perp e^{-i\mathbf{k}_{1\perp}\cdot\mathbf{r}_{xx'}-i\mathbf{k}_{2\perp}\cdot\mathbf{r}_{yy'}} (4\alpha_s C_F) \Xi_{\text{LO}} \\ &\times \frac{e^{-i\mathbf{k}_{g\perp}\cdot\mathbf{r}_{xx'}}}{(\mathbf{k}_{g\perp} - \frac{z_g}{z_1}\mathbf{k}_{1\perp})^2} \left\{ 8z_1z_2^3(1-z_2)^2Q^2\left(1 + \frac{z_g}{z_1} + \frac{z_g^2}{2z_1^2}\right) K_0(\bar{Q}_{\text{R2}}r_{xy})K_0(\bar{Q}_{\text{R2}}r_{x'y'})\delta_z^{(3)} \right. \\ &\left. - \mathcal{R}_{\text{LO}}^{\text{L}}(\mathbf{r}_{xy}, \mathbf{r}_{x'y'})\Theta(z_1 - z_g)\delta_z^{(2)} \right\} + (1 \leftrightarrow 2), \quad \bar{Q}_{\text{R2}}^2 = z_2(1-z_2)Q^2. \end{aligned}$$

The phase-space integration must exclude the domain in which the quark and the gluon lie inside the same jet — that region is already in $d\sigma_{\text{V,no-sud,LO}}$. This is precisely the out-of-cone condition of §4.1.

(IV-b) Finite real, gluon crossing the shockwave — CSSV (3.23)

Depends on $\Xi_{\text{NLO},1}$ and $\Xi_{\text{NLO},4}$, on $X_{\text{R}}^2 = z_1z_2r_{xy}^2 + z_1z_g\mathbf{r}_{zx}^2 + z_2z_g\mathbf{r}_{zy}^2$ (3.25) and on $\mathbf{w}_\perp = (z_1\mathbf{x}_\perp + z_g\mathbf{z}_\perp)/(z_1 + z_g)$ (3.26). Its last term, $-\frac{\alpha_{\text{em}}e_f^2N_c\delta_z^{(2)}}{(2\pi)^8}\alpha_s\Theta(z_f - z_g) \times$ "slow", is the $z_g \rightarrow 0$ limit of the formula itself, subtracted so that the gluon-integrated cross-section is finite as $\Lambda^- \rightarrow 0$; the corresponding counterterm is what appears in piece (I). **This is the trijet's own rapidity subtraction and it is exactly what you asked to construct.**

8. Pitfalls, in decreasing order of how often they bite

1. **Over-subtracting the rapidity divergence.** Slicing at $z_g < z_f$ over the whole transverse plane hands the Sudakov region to the evolution. Symptom: wrong-sign Sudakov double log. Cure: §5.
2. **Double counting the lifetime correction.** Δ_{lt} is the difference between constrained and unconstrained evolution. Add it to the impact factor *or* use the constrained kernel — never both (§6.2, step 5).
3. **Trying to build BK/JIMWLK from the real trijet alone.** The virtual $q\bar{q}$ diagrams supply the $U^\dagger U$ (virtual) terms of η^{ab} . Without them the $\mathbf{z}_\perp$ integral of the kernel does not converge and the evolution is not probability conserving (§2.1).
4. **Confusing $\ln(1/z_0)$ with $\ln^2(1/z_0)$.** Only the single log is factorized into the target. A $\ln^2$ that refuses to cancel means the in-cone/out-of-cone or the real/virtual bookkeeping is inconsistent.
5. **Dropping z_g in the eikonal denominator.** In $1/(\mathbf{k}_{g\perp} - \frac{z_g}{z_1}\mathbf{k}_{1\perp})^2$ it is tempting to set $z_g = 0$. In back-to-back kinematics $\mathbf{k}_{g\perp} \sim \mathbf{q}_\perp$ and $\frac{z_g}{z_1}\mathbf{k}_{1\perp} \sim z_g\mathbf{P}_\perp$ are the *same order* even for small z_g ; the term must be kept [CSSV footnote 31]. It is only droppable if you want the leading $z_g \rightarrow 0$ divergence and nothing else.
6. **Choosing the jet algorithm after the fact.** The in-cone weight, the $\ln R$ coefficient and the finite constants all depend on the clustering criterion. CSSV use the cone/E-scheme criterion (3.2)–(3.4); the generalized- k_t family gives (3.5) and different finite terms (CSSV Appendix D). In the small- R limit they differ only by $\mathcal{O}(\alpha_s)$ constants — but those constants matter at NLO.
7. **Forgetting that R -dependence does not cancel.** Unlike the slicing parameter δ , R is part of the observable definition. Only $\mathcal{O}(R^2)$ is dropped.
8. **Setting the running-coupling scale by hand and claiming NLO accuracy.** The one-loop correction to the classical shockwave background contains the β -function and turns α_s into a running coupling, but the *scale* can only be fixed at two loops, i.e. by the NNLO impact factor [CSSV footnote 10, and Eqs. (4.58)–(4.59)].
9. **Large- N_c too early.** The kernel structure at finite N_c requires quadrupoles ($\Xi_{\text{NLO},1,2,4}$). Take $N_c \rightarrow \infty$ only after the divergence extraction, not before.
10. **Assuming TMD factorization survives.** CSSV argue it holds only in the window $Q_s \ll q_\perp \ll P_\perp$; when $Q_s \sim q_\perp \ll P_\perp$ it is violated at NLO by higher twists. That is a statement about the finite pieces, not about the divergence extraction, but it limits what you may do with the answer.

Appendix A. The over-subtraction integral, in full

We evaluate

$$\Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi^2} \hat{G}_{Y_f}^{ij}(\mathbf{b}_\perp, \mathbf{b}'_\perp) \int_{z_0}^{z_f} \frac{dz_g}{z_g} \int d^2 \mathbf{z}_\perp \Theta\left(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2 - \frac{z_f}{z_g}\right) \frac{\mathbf{r}_{bb'}^2}{\mathbf{r}_{zb}^2 \mathbf{r}_{zb'}^2},$$

i.e. CSSV (4.50): the piece of phase space that a naive $z_g < z_f$ slice would hand to the evolution but that lifetime ordering forbids.

Step 1: the longitudinal integral

Substitute $t = \ln(z_f/z_g)$, so that $dz_g/z_g = dt$ and $z_g \in (z_0, z_f) \Leftrightarrow t \in (0, T)$ with $T = \ln(z_f/z_0)$. Writing

$$\rho \equiv \ln(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2),$$

the step function is simply $\Theta(\rho - t)$, so

$$\int_{z_0}^{z_f} \frac{dz_g}{z_g} \Theta(\dots) = \int_0^T dt \Theta(\rho - t) = \min(\rho, T) \xrightarrow{T \rightarrow \infty} \rho \Theta(\rho).$$

Note that z_0 has dropped out. The lifetime constraint provides its own cutoff on the z_g integration: a gluon can only violate lifetime ordering if $\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2 > 1$, and then only down to $z_g > z_f/(\min(\dots) Q_f^2)$. This is why Δ_{lt} is a genuinely finite $\mathcal{O}(\alpha_s)$ correction to the impact factor, not a redefinition of the divergence.

$$\Rightarrow \Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi^2} \hat{G}_{Y_f}^{ij} \int d^2 \mathbf{z}_\perp \frac{\mathbf{r}_{bb'}^2}{\mathbf{r}_{zb}^2 \mathbf{r}_{zb'}^2} \ln(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2) \Theta(\min(\mathbf{r}_{zb}^2, \mathbf{r}_{zb'}^2) Q_f^2 - 1).$$

Step 2: the transverse integral

The dipole kernel $\mathbf{r}_{bb'}^2/(\mathbf{r}_{zb}^2 \mathbf{r}_{zb'}^2)$ is logarithmically divergent only at the two endpoints $\mathbf{z}_\perp \rightarrow \mathbf{b}_\perp$ and $\mathbf{z}_\perp \rightarrow \mathbf{b}'_\perp$; at large $|\mathbf{z}_\perp|$ it falls as $\mathbf{r}_{bb'}^2/|\mathbf{z}_\perp|^4$ and, even multiplied by $\ln|\mathbf{z}_\perp|^2$, converges. The double logarithm therefore comes entirely from the two endpoint regions.

Take $\mathbf{z}_\perp \rightarrow \mathbf{b}_\perp$ and set $\mathbf{r} \equiv |\mathbf{r}_{zb}| \ll |\mathbf{r}_{bb'}|$. Then $|\mathbf{r}_{zb'}| \simeq |\mathbf{r}_{bb'}|$, $\min(\dots) = r^2$, and the kernel reduces to $1/r^2$. The two Θ -conditions become $1/Q_f < r$, while the endpoint approximation holds for $r < |\mathbf{r}_{bb'}|$:

$$\int d^2 \mathbf{z}_\perp \frac{\ln(r^2 Q_f^2)}{r^2} = 2\pi \int_{1/Q_f}^{|\mathbf{r}_{bb'}|} \frac{dr}{r} \ln(r^2 Q_f^2) \stackrel{s=\ln(r^2 Q_f^2)}{=} 2\pi \int_0^{\rho_{\max}} \frac{ds}{2} s = \frac{\pi}{2} \rho_{\max}^2, \quad \rho_{\max} \equiv \ln(\mathbf{r}_{bb'}^2 Q_f^2).$$

The endpoint $\mathbf{z}_\perp \rightarrow \mathbf{b}'_\perp$ gives an identical contribution, so the total is $\pi \rho_{\max}^2$. Hence

$$\Delta_{\text{lt}} = -\frac{\alpha_s N_c}{2\pi^2} \hat{G}_{Y_f}^{ij} \times \pi \rho_{\max}^2 = -\frac{\alpha_s N_c}{2\pi} \ln^2(Q_f^2 \mathbf{r}_{bb'}^2) \hat{G}_{Y_f}^{ij},$$

which is CSSV (4.51). (CSSV evaluate the integral exactly in their Appendix F and find that the remainder is a power correction, $\mathcal{O}(1/(\mathbf{r}_{bb'}^2 Q_f^2))$, with no leftover single logarithm.)

Geometric reading. Since $\int d^2 \mathbf{z}_\perp / \mathbf{r}_{zb}^2 = \pi \int d\rho$ and the longitudinal integral contributed the factor ρ (the height of the triangle at abscissa ρ),

$$\int d^2 \mathbf{z}_\perp \int \frac{dz_g}{z_g} \Big|_{\text{over-sub}} = \pi \int_0^{\rho_{\max}} \rho d\rho = \pi \cdot \underbrace{\frac{1}{2} \rho_{\max}^2}_{\text{area of the triangle, Fig. 2}}$$

The Sudakov double logarithm is the area of the region between the naive rapidity-slicing boundary and the lifetime-ordering boundary. That is the cleanest one-line summary of why the two uploaded papers belong together.

Appendix B. Glossary of symbols

Symbol	Meaning
z_1, z_2, z_g	k^-/q^- fractions of quark, antiquark, gluon; $z_1 + z_2 + z_g = 1$ (trijet)
$\mathbf{x}_\perp, \mathbf{y}_\perp, \mathbf{z}_\perp$	transverse positions of $q, \bar{q}, g$ at the shockwave (primed: conjugate amplitude)
$\mathbf{r}_{xy}$	$\mathbf{x}_\perp - \mathbf{y}_\perp$
$\mathbf{u}_\perp, \mathbf{b}_\perp$	relative and "centre-of-mass" dipole coordinates; conjugate to $\mathbf{P}_\perp, \mathbf{q}_\perp$
$\bar{Q}^2$	$z_1 z_2 Q^2$
X_R, X_V	effective $q\bar{q}g$ dipole sizes, real (3.25) and virtual (3.20)
$\mathcal{C}_{ik\perp}$	collinearity variable of the jet algorithm, CSSV (3.3)
R	jet radius; the collinear slicing parameter
δ	soft slicing parameter on z_g (analogue of HO's δ_s)
$\Lambda^-, z_0 = \Lambda^-/q^-$	rapidity regulator
$k_f^-, z_f = k_f^-/q^-, Y_f = \ln(1/z_f)$	rapidity factorization scale
$k_f^+, Q_f^2 = 2k_f^+ k_f^-$	conjugate plus-momentum scale; sets the lifetime constraint, $Q_f^2 \sim \mathbf{P}_\perp^2$
$D_{xy}, Q_{xy,y'/x'}$	dipole, quadrupole Wilson-line operators
$\Xi_{\text{LO}}, \Xi_{\text{NLO},1..4}$	colour correlators, CSSV Table 1
$\mathcal{H}_{\text{LL}}, \eta^{ab}$	leading-log JIMWLK Hamiltonian and its kernel, CSSV (3.30)–(3.31)
$\hat{G}_Y^{ij}(\mathbf{b}_\perp, \mathbf{b}'_\perp)$	coordinate-space Weizsäcker–Williams gluon TMD
c_0	$2e^{-\gamma_E}$
$\tilde{\mu}^2$	$4\pi e^{-\gamma_E} \mu^2$, the $\overline{\text{MS}}$ transverse scale
S, HC, $\overline{\text{HC}}$	soft / hard-collinear / hard-non-collinear slices of the gluon phase space

References

1. **[CSSV]** P. Caucal, F. Salazar, B. Schenke, R. Venugopalan, *Back-to-back inclusive dijets in DIS at small x : Sudakov suppression and gluon saturation at NLO*, JHEP (2022), arXiv:2208.13872v2 [hep-ph]. — source of all CGC conventions, the NLO decomposition (3.6), the JIMWLK factorization §3.3, and the lifetime-ordering analysis §4.3.
2. **[HO]** B. W. Harris, J. F. Owens, *The two cutoff phase space slicing method*, Phys. Rev. D **65** (2002) 094032. — source of the slicing machinery: soft region and eikonal approximation (§II A), collinear region and splitting functions (§II B), factorization of initial/final-state singularities (§II B 1–3), cutoff cancellation (§III, Appendix A) and numerical-convergence tricks (Appendix D).
3. P. Caucal, F. Salazar, R. Venugopalan, *Dijet impact factor in DIS at NLO in the CGC*, JHEP 11 (2021) 222, arXiv:2108.06347 — the original NLO dijet impact factor that CSSV reorganize; contains the full $q\bar{q}g$ expressions and the function $\mathcal{J}_\odot$.

Scope and honesty note. §§1–2 and §§4–6 are an organization, in the Harris–Owens slicing language, of results that exist in the literature; every explicit analytic weight quoted here is taken from CSSV with its equation number given, and none has been re-derived independently except the integral in Appendix A, which is done in full above and agrees with CSSV (4.51). Where a statement is a structural claim about the method rather than a quoted result — the region definitions of §4.1, the dictionary of §3.2, the algorithm of §6.2 — it is presented as such. Numerical implementation of the trijet Monte Carlo (piece IV) is left open; CSSV likewise leave the three-body phase-space integration to future work.